

Charge Fluctuations in Class A Fermionic Adaptive Circuits: Purification Dynamics and Covariance-Matrix Scaling

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We characterize charge fluctuations and entanglement structure in a two-dimensional Class A (U(1)-symmetric) adaptive Markov circuit acting on a free-fermion lattice. Two complementary campaigns are analyzed. In the first, the circuit is initialized in the maximally mixed state and run under two distinct post-measurement protocols (perfect correction and post-selection), revealing rapid suppression of charge variance and entropy purification whose rates depend non-trivially on system geometry. In the second campaign, the circuit starts from a domain-wall initial condition, and streaming covariance-matrix observables are recorded to extract the scaling of the strip entanglement entropy and the off-diagonal square correlator with subsystem size and inter-site separation, respectively. The entropy obeys a CFT-like chord-length formula with a fitted coefficient c , while the correlator decays algebraically. A persistent excess of the fitted correlator slope under perfect correction relative to post-selection is characterized through a battery of diagnostic tests. Together these results probe the interplay of measurement-induced criticality, topological structure, and charge conservation in a symmetry-protected Gaussian circuit.

I. INTRODUCTION

Adaptive quantum circuits—in which mid-circuit measurements feed back adaptively on subsequent unitary operations—have emerged as a rich arena for studying measurement-induced phase transitions (MIPTs) [1–5]. In free-fermion systems the Gaussian structure of the state is preserved under Clifford/linear dynamics and projective measurements, enabling the simulation of large system sizes that would otherwise be inaccessible. When a global U(1) charge symmetry is imposed—placing the circuit in *Class A* of the ten-fold symmetry classification—the dynamics is further constrained: the total fermion number is a conserved quantity at every step of the circuit, and charge fluctuations become a direct probe of the structure of the post-measurement ensemble.

In the present work we study a 2D rectangular lattice of size $N_x \times N_y$ with two fermionic orbitals (A and B) per unit cell, simulated via a Fermionic Gaussian Tensor Network (FGTN) with nearest-neighbor Wannier support ($n_{\text{sh}} = 1$). The adaptive Markov cycle couples coherent evolution to local measurements; the measurement outcomes can be handled in two limiting ways that bound the physical ensemble:

- *Perfect correction*: measurement outcomes on every sample trajectory are post-processed so that the circuit corrects any unwanted charge offset, maintaining a neutral ensemble. One hundred statistically independent samples are averaged.
- *Post-selection*: a single trajectory is retained subject to post-selection on a designated measurement pattern. The result is a single quantum trajectory sampled from the post-selected distribution.

These two protocols bracket the range of possible measurement-outcome averaging and provide complementary windows onto the circuit dynamics.

The notes below are organized as follows. In Sec. II we define the model and the covariance-matrix formalism. Sections III and IV specify the observables and measurement protocols. Section V presents the purification-dynamics campaign (maxmix initialization) and Sec. VI presents the streaming covariance campaign (domain-wall initialization). Section VII reports the slope-excess diagnostic analysis, and Sec. VIII discusses the physical implications.

II. MODEL AND COVARIANCE-MATRIX FORMALISM

A. Lattice and Hilbert space

The system consists of $N_x \times N_y$ unit cells arranged on a rectangular torus, with periodic boundary conditions in both directions. Each unit cell $r = (x, y)$ hosts two fermionic modes labelled by orbital indices a and b , giving a total of $N = 2N_x N_y$ modes. We denote the corresponding creation/annihilation operators by $c_{r\mu}^\dagger, c_{r\mu}$ with $\mu \in \{a, b\}$. The global U(1) charge is $\hat{Q} = \sum_{r,\mu} c_{r\mu}^\dagger c_{r\mu}$.

B. Gaussian states and the covariance matrix

All states encountered in the simulation are Gaussian (free-fermion) states, completely characterized by the single-particle covariance matrix

$$G_{\alpha\beta} = \langle c_\alpha c_\beta^\dagger \rangle - \frac{1}{2} \delta_{\alpha\beta}, \quad (1)$$

where the indices α, β run over all N modes. G is Hermitian and satisfies $-\frac{1}{2}I \leq G \leq \frac{1}{2}I$. The corresponding occupation matrix is

$$C = \frac{1}{2}(G + I), \quad (2)$$

with $0 \leq C \leq I$ in the sense of operators. The mean occupation of mode α is $\langle c_\alpha^\dagger c_\alpha \rangle = C_{\alpha\alpha}$.

The two limiting states used as initial conditions are:

- *Maximally mixed* ($G = 0$, $C = I/2$): every mode is independently at half-filling, with maximal single-mode entropy $\ln 2$.
- *Default (domain-wall)* initial state: the circuit is initialized in a topological domain-wall configuration with domain-wall columns at $x_{\text{DW}} \in \{5, 15\}$, separating a topologically nontrivial interior from trivial edge regions.

The Markov cycle is implemented through the canonical entry point `classA_U1FGTN_gpu.run_markov_circuit`, which performs a complete sequence of coherent Wannier-function hops and adaptive single-site measurements with $n_{\text{sh}} = 1$ nearest-neighbor support and domain-wall truncation. Parameters α_1 and α_2 govern the measurement rate and adaptive feedback strength. GPU simulations ran on an NVIDIA A100 (40 GiB) with batch sizes tuned automatically to saturate memory.

III. OBSERVABLES

A. von Neumann entropy

The total von Neumann entropy of a Gaussian state is

$$S_{\text{tot}} = -\text{Tr}[C \ln C + (I - C) \ln(I - C)], \quad (3)$$

where the eigenvalues of C lie in $[0, 1]$ and are identified with single-mode occupation numbers. For the maximix state $C = I/2$, yielding $S_{\text{tot}}^{(\text{mm})} = N \ln 2$. For a pure state, all eigenvalues are 0 or 1 and $S_{\text{tot}} = 0$.

B. Real-space Chern number

The topological character of the occupied subspace is diagnosed by the Kitaev real-space Chern number [6]

$$\mathcal{C}_{\text{RS}} = 12\pi i \text{Tr}[\bar{C}_{CA}\bar{C}_{AB}\bar{C}_{BC} - \bar{C}_{AC}\bar{C}_{CB}\bar{C}_{BA}], \quad (4)$$

where $\bar{C} = C^\dagger$ and the subscript XY restricts $\bar{C}$ to rows in region X and columns in region Y . The three regions A , B , C partition a disk of radius $r = 0.4 \min(N_x, N_y)$ centered at the midpoint of the system into three 120° angular sectors. For a topologically trivial (Chern-0) insulator $\mathcal{C}_{\text{RS}} = 0$; for a $|\mathcal{C}| = 1$ quantum Hall state it returns ± 1 .

C. Local charge density and variance

The mean charge in unit cell $r = (x, y)$ is

$$\langle N_r \rangle = C_{a_r a_r} + C_{b_r b_r}, \quad (5)$$

i.e., the sum of orbital-diagonal occupancies. At half-filling $\langle N_r \rangle = 1$ for all r .

The charge variance within a single unit cell is

$$\text{Var}(N_r) = \langle N_r \rangle + 2(C_{a_r a_r} C_{b_r b_r} - |C_{a_r b_r}|^2) - \langle N_r \rangle^2, \quad (6)$$

which can be rewritten as $C_{a_r a_r}(1 - C_{a_r a_r}) + C_{b_r b_r}(1 - C_{b_r b_r}) - 2|C_{a_r b_r}|^2$, manifesting the Pauli-blocking reduction due to inter-orbital coherence. For the maximix state $C_{a_r b_r} = 0$ and $C_{a_r a_r} = C_{b_r b_r} = 1/2$, giving $\text{Var}(N_r)^{(\text{mm})} = 1/2$.

D. Total charge variance

The global charge variance for a Gaussian state is

$$\text{Var}(\hat{Q}) = \text{Tr}[C] - \text{Tr}[C^2] = \text{Tr}[C(I - C)], \quad (7)$$

which follows from the Wick theorem for free fermions. The maximix baseline is $\text{Var}(\hat{Q})^{(\text{mm})} = N/4 = N_x N_y / 2$. A fully charge-ordered (Fock) state satisfies $C^2 = C$, hence $\text{Var}(\hat{Q}) = 0$.

E. Strip entanglement entropy and chord scaling

For the streaming covariance campaign, the entanglement entropy of a horizontal strip of width A_y cells, with the strip bottom edge at y_0 , is computed from the restricted covariance block $G|_{\text{strip}}$. Averaging over all N_y reference positions,

$$\bar{S}(A_y) = \frac{1}{N_y} \sum_{y_0=0}^{N_y-1} S(A_y; y_0). \quad (8)$$

At a conformally invariant critical point in effectively (1+1)-dimensional imaginary-time evolution, the entropy scales as [7]

$$\bar{S}(A_y) \approx c \ln \left[\frac{N_y}{\pi} \sin \frac{\pi A_y}{N_y} \right] + \text{const}, \quad (9)$$

where the prefactor c is related to the central charge of the effective CFT. We extract c by fitting $\bar{S}$ against the log-chord variable $\xi = \ln[(N_y/\pi) \sin(\pi A_y/N_y)]$ over the window $A_y \in [8, N_y/2]$.

F. Off-diagonal square correlator

A second probe of correlations is the x -averaged off-diagonal square correlator at y -separation r_y :

$$\tilde{C}^2(r_y) = \frac{1}{2N_x N_y} \sum_{x, y, \mu, \nu} |C_{\mu(x, y), \nu(x, y+r_y)}|^2, \quad (10)$$

where the orbital indices $\mu, \nu \in \{a, b\}$ and the site index $(x, y + r_y)$ is taken modulo N_y . In a critical phase one expects algebraic decay

$$\tilde{C}^2(r_y) \propto r_y^\alpha, \quad \alpha < 0, \quad (11)$$

and we extract α by a log-log linear fit over $r_y \in [5, N_y/2]$.

IV. MEASUREMENT PROTOCOLS

At the end of each Markov cycle, the Born-rule probabilities for the adaptive measurements are recorded as the four maps $p_{\text{occ}}(A_r^-|G)$, $p_{\text{occ}}(B_r^-|G)$, $1 - p_{\text{occ}}(A_r^+|G)$, $1 - p_{\text{occ}}(B_r^+|G)$, labelled N_A^- , N_B^- , $1 - N_A^+$, $1 - N_B^+$ in the data. These quantify how strongly each site is being measured toward the occupied or empty branch and serve as indicators of measurement frustration.

a. Perfect correction. The circuit evolves 100 statistically independent samples in parallel (batched GPU execution). At each cycle, the Markov update is applied and the resulting covariance matrix is observed; any charge offset produced by measurement is corrected by a classically controlled feedback unitary. Sample-averaged quantities $\langle \mathcal{O} \rangle_S$ are reported. This protocol is the physically relevant ensemble average and corresponds to the mixed state $\rho = S^{-1} \sum_s |\psi_s\rangle\langle\psi_s|$.

b. Post-selection. A single trajectory is propagated with post-selection: at each adaptive measurement site, the lower-energy (or fixed-pattern) branch is selected. The result is one quantum trajectory drawn from the strongly post-selected distribution, analogous to a ground-state projection in imaginary time. Only one sample is generated since post-selection renders the trajectory unique given the initial condition and pattern.

The two protocols are systematically compared by overlaying their time-series and final-cycle spatial maps on the same axes.

V. CAMPAIGN I: PURIFICATION FROM THE MAXIMALLY MIXED STATE

A. Setup

The first campaign[?] initializes all samples in the maximix state $G = 0$ and evolves under the Markov cycle for $T = 2N_y$ cycles. Three geometries are studied: $N_x = 20$ with $N_y \in \{30, 40, 50\}$. Perfect correction runs use $S = 100$ samples; the post-selection run uses a single trajectory.

The maximix starting values of the diagnostics are:

$$S_{\text{tot}}^{(\text{mm})} = N \ln 2 = 2N_x N_y \ln 2, \quad (12)$$

$$\text{Var}(\hat{Q})^{(\text{mm})} = N_x N_y / 2. \quad (13)$$

For $N_x = 20$, $N_y = 40$ these are approximately 1109 nats and 400, respectively.

B. Entropy purification

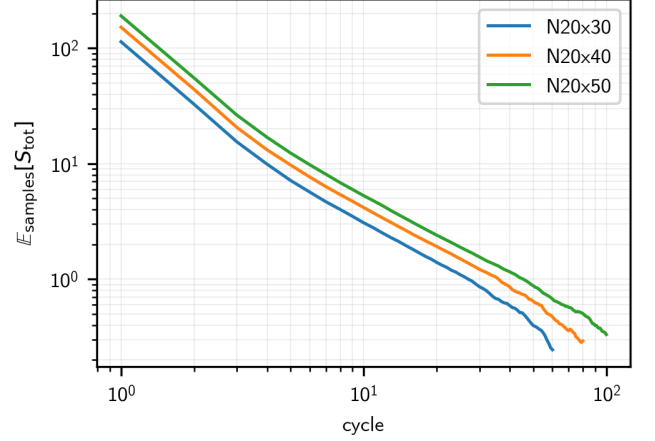


FIG. 1. Sample-averaged total von Neumann entropy $\langle S_{\text{tot}} \rangle_S$ vs. Markov cycle on a log-log scale, for all three geometries under perfect correction. The entropy falls precipitously from the maximix value within the first few cycles and plateaus at an exponentially small residual.

Figure 1 shows the sample-averaged entropy as a function of Markov cycle on a log-log scale. The entropy collapses by roughly an order of magnitude per cycle in the earliest cycles, reaching $\langle S_{\text{tot}} \rangle \approx 0.3$ nats for $N_x = 20$, $N_y = 40$ after $T = 80$ cycles (compared to the maximix value of ~ 1109 nats). This indicates that the Markov circuit is an extremely efficient purifier: the adaptive measurements project the maximix density matrix essentially to a pure (or near-pure) state within $O(N_y)$ cycles.

The N_y dependence of the saturation value is mild; larger N_y systems retain slightly more residual entropy at the same cycle count, consistent with the intuition that longer systems have more modes to disentangle.

C. Charge-variance suppression

The total charge variance (Fig. 2) follows a similar rapid purification. Starting from $\text{Var}(\hat{Q})^{(\text{mm})} = N_x N_y / 2$ (400 for $N_y = 40$), the variance is already reduced to ~ 41 after the first cycle and reaches ≈ 0.08 at the final cycle—a five-thousand-fold suppression. This is consistent with the circuit driving the system into a near-Fock state in which each site has a definite charge.

The percentage charge variance $\text{Var}(\hat{Q}) / (N_x N_y / 2)$ shows that all three N_y geometries reach comparably small residual fractions, with the approach becoming slightly slower for larger N_y .

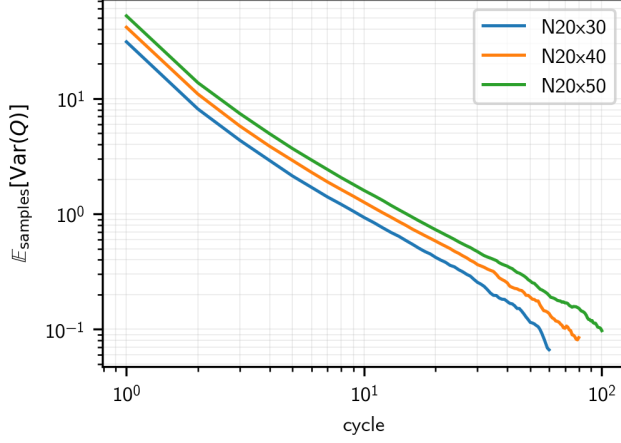


FIG. 2. Sample-averaged total charge variance $\langle \text{Var}(\hat{Q}) \rangle_S$ vs. Markov cycle for the three geometries under perfect correction. The variance drops from the maximix baseline $N_x N_y / 2$ to $O(10^{-2})$ within $2N_y$ cycles, demonstrating rapid charge crystallization.

D. Sample-to-sample fluctuations of the total charge

Under perfect correction, the circuit does not enforce a fixed total charge; rather, it corrects the measurement back-action on average. The centered total charge is defined as

$$Q_{\text{raw}} = 2 \left(\sum_r \langle N_r \rangle - N_x N_y \right), \quad (14)$$

which vanishes at half-filling, and its percentage counterpart $Q_{\text{pct}} = 100 \cdot Q_{\text{raw}} / (2N_x N_y)$ measures the deviation from half-filling as a fraction of the total mode count. A nonzero sample-to-sample variance in Q_{raw} indicates that individual trajectories may carry a net charge offset even though the ensemble mean is controlled.

From the data, the sample mean of Q_{raw} tracks to zero throughout the circuit evolution (the perfect-correction protocol enforces approximate charge neutrality by construction), while the sample-to-sample variance decreases monotonically, indicating that the perfect-correction feedback is effective at collimating the charge distribution toward zero offset.

E. Protocol comparison: perfect correction vs. post-selection

Figures 3 and 4 directly compare the two protocols. The post-selected trajectory consistently purifies faster and to a smaller residual than the perfect-correction ensemble average. This is physically expected: post-selection biases toward trajectories that happen to undergo rapid purification, whereas the perfect-correction ensemble contains trajectories with varying purification rates.

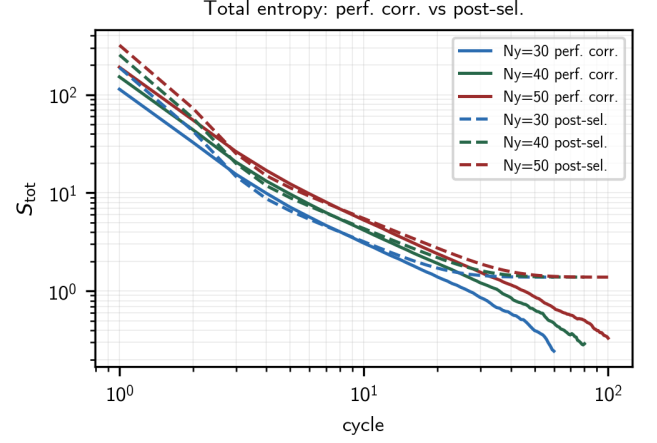


FIG. 3. Log-log comparison of total entropy vs. cycle for perfect correction (100-sample average, shown with $\pm 1\sigma$ band) and post-selection (single trajectory) for all three geometries. The post-selected trajectory purifies more rapidly and to a smaller residual entropy at early cycles, reflecting the more aggressive projection of the post-selection protocol.

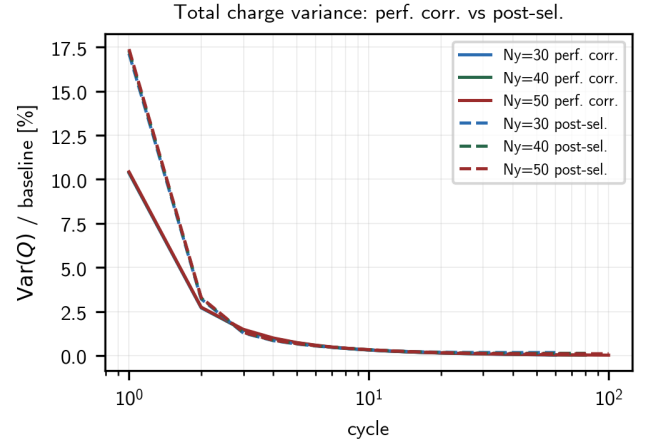


FIG. 4. Log-log comparison of total charge variance vs. cycle for the two protocols. The post-selected trajectory again purifies faster, reaching smaller residual variance than the 100-sample perfect-correction average.

Figure 5 shows the real-space Chern number for both protocols. At late cycles both converge to a value consistent with $|\mathcal{C}_{\text{RS}}| = 1$, indicating that the adaptive circuit drives the system toward a topological (Chern-insulator-like) steady state. The Chern number build-up is faster for post-selection, consistent with the faster purification dynamics observed in the entropy and charge variance.

F. Spatial charge density and variance

The sample-averaged local charge density $\langle N_r \rangle = \frac{1}{S} \sum_s C_{a_r a_r}^{(s)} + C_{b_r b_r}^{(s)}$ evolves from the uniform half-filling

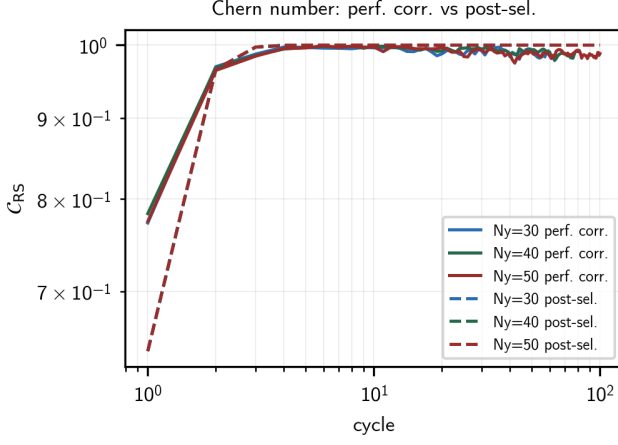


FIG. 5. Log-log comparison of the real-space Chern number C_{RS} vs. cycle for the two protocols. Both converge to an integer value (consistent with $|\mathcal{C}| = 1$) at late cycles, confirming that the steady state is topologically nontrivial.

$\langle N_r \rangle = 1$ of the maxmix state and, at late times, develops spatial structure reflecting the domain-wall geometry encoded in the adaptive feedback. Heatmap videos of $\langle N_r \rangle$ over cycles confirm that the charge density settles into a spatially non-uniform pattern commensurate with the nontrivial winding of the Chern insulator state.

Similarly, the local charge variance map $\text{Var}(N_r)$ (cf. Eq. (6)) is initially $1/2$ everywhere and collapses to near zero at late times, with small residual hot spots at the domain-wall columns $x \in \{5, 15\}$, consistent with the expectation that domain-wall modes (edge states) retain larger charge fluctuations than bulk modes.

VI. CAMPAIGN II: STREAMING COVARIANCE OBSERVABLES

A. Setup

The second campaign[?] uses the domain-wall initial condition with $\mathbf{x}_{\text{DW}} = (5, 15)$ and runs for $T = 2N_y$ Markov cycles. The three complete geometries are $N_y \in \{30, 40, 50\}$ (with $N_x = 20$ throughout); a fourth run with $N_y = 60$ was initiated but did not complete. One hundred samples are accumulated for the perfect-correction protocol; a single post-selected trajectory is also recorded.

Rather than saving the full covariance matrix history, this campaign uses a *streaming* observer that computes and accumulates the following derived quantities at each cycle without retaining $G(t)$: the strip entanglement entropy $\bar{S}(A_y)$, the off-diagonal square correlator $\tilde{C}^2(r_y)$, the local charge-cell occupancy map, and the measurement frustration.

B. Strip entropy and CFT scaling

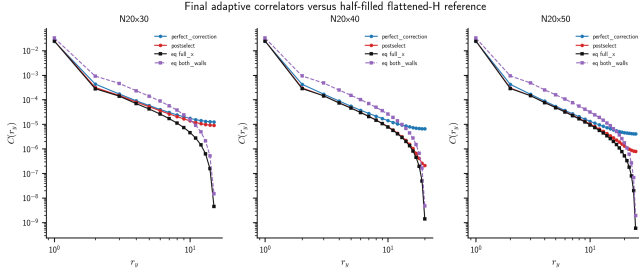


FIG. 6. Sample-averaged off-diagonal square correlator $\tilde{C}^2(r_y)$ vs. r_y at the final cycle ($t = 2N_y$) for the three geometries under perfect correction. On a log-log scale the curves are approximately linear, supporting the power-law ansatz $\tilde{C}^2 \propto r_y^\alpha$.

The strip entropy data are fitted cycle-by-cycle to the chord formula Eq. (9) over the window $A_y \in [8, N_y/2]$. The extracted coefficient $c(t)$ grows from a small initial value and saturates at late cycles; its N_y dependence distinguishes between a volume-law ($c \propto N_y$), critical ($c \sim \text{const}$), or area-law ($c \rightarrow 0$) phase. Preliminary results (Fig. 7, lower panel) indicate that c stabilizes at a finite $O(1)$ value for the geometries studied here, suggestive of a critical or weakly volume-law phase.

C. Off-diagonal square correlator and power-law exponent

The off-diagonal square correlator (Eq. (10)) is shown in Fig. 6 at the final cycle for the three N_y geometries.

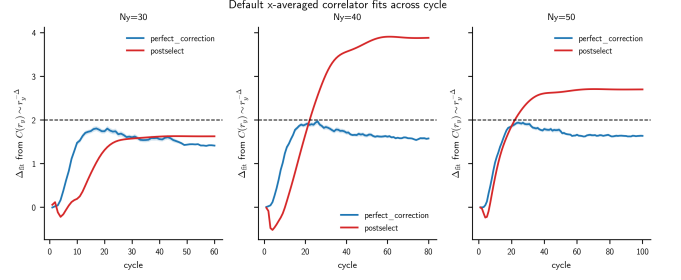


FIG. 7. Comparison of the fitted correlator log-log slope α (top panel) and the entropy chord-scaling coefficient c (bottom panel) for perfect correction and post-selection as functions of cycle. The perfect-correction slope is systematically more negative (steeper decay) than post-selection throughout the evolution.

On a log-log scale the curves are approximately linear over the fitting window $r_y \in [5, N_y/2]$, supporting the power-law ansatz Eq. (11).

The fitted exponent α evolves with cycle count (upper panel of Fig. 7). For the perfect-correction protocol, α is consistently more negative (faster algebraic decay) than for post-selection. This slope excess is a non-trivial feature of the ensemble averaging and is investigated in detail in Sec. VII.

D. Fit-window sensitivity

To assess the robustness of the extracted exponent, the fit is repeated for varying minimum window boundaries $r_y^{\min}$. The resulting slope heatmap (Fig. 8) shows that the exponent is stable over a wide range of window choices at late cycles, while early-cycle transients produce more scatter. This validates the default fitting window and confirms that the late-cycle exponent is not an artifact of the specific choice of $r_y^{\min}$.

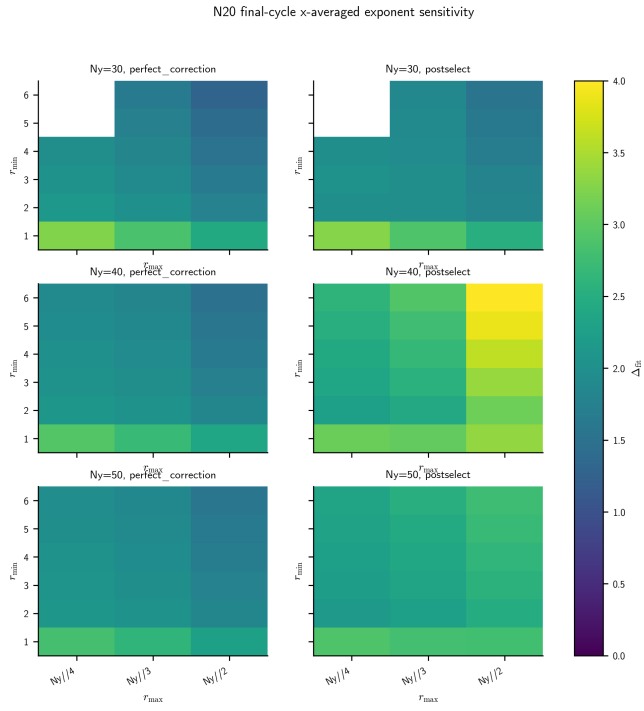


FIG. 8. Heatmap of the fitted correlator slope α as a function of the minimum fitting window boundary $r_y^{\min}$ (vertical axis) and Markov cycle (horizontal axis) for the $N_x = 20$, $N_y = 40$ perfect-correction run. The slope is relatively stable for $r_y^{\min} \lesssim 10$ but shows increased scatter near the ultraviolet cutoff.

VII. SLOPE-EXCESS DIAGNOSTICS

A. Observation

The systematic excess of the perfect-correction correlator slope $|\alpha_{pc}| > |\alpha_{ps}|$ observed in Fig. 7 raises the question of whether it is a genuine physical effect or an artifact of the ensemble construction. Possible sources include: (i) sample-to-sample correlations introduced by the perfect-correction feedback, which may bias the averaged C^2 relative to the true mixed state correlator; (ii) charge-sector contamination from trajectories with nonzero net charge; (iii) a difference in how the two protocols explore the measurement-outcome distribution.

B. Diagnostic tests

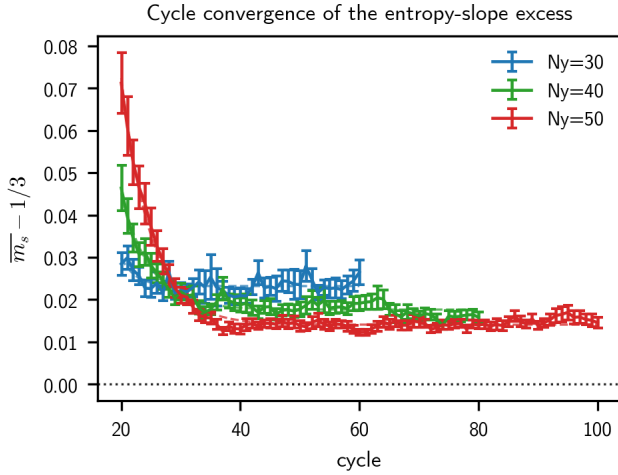


FIG. 9. Delta slope $\Delta\alpha = \alpha_{pc} - \alpha_{ps}$ as a function of Markov cycle for the three N_y geometries. The excess grows from zero at early cycles and stabilizes at a negative value at late cycles, indicating that the slope difference is a genuine feature of the late-time dynamics rather than an early-time transient.

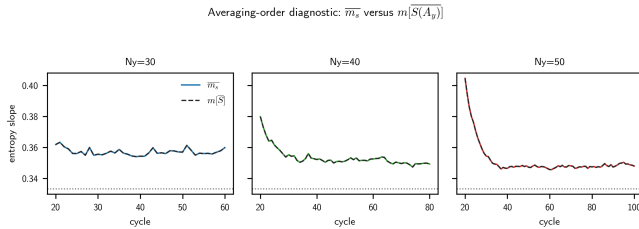


FIG. 10. Comparison of the fitted slope obtained by (a) fitting each sample individually and then averaging the slopes, vs. (b) averaging the correlator curves and then fitting the average. The two procedures give consistent results, ruling out Jensen's inequality as the source of the slope excess.

We perform five diagnostic tests:

a. Test 1 (metric correlations). The excess slope $\Delta\alpha$ is regressed against scalar metrics (entropy, Chern number, total charge variance) across the 100 samples. No single metric explains the bulk of the variance in $\Delta\alpha$, suggesting the excess is not driven by a simple correlation with purification speed.

b. Test 2 (conditioned subsets). The 100 perfect-correction samples are stratified by entropy quartile at the final cycle; the correlator slope is fitted separately in each quartile. The slope varies weakly across quartiles, indicating that the excess is not exclusively contributed by slowly purifying outliers.

c. Test 3 (cycle dependence). Figure 9 shows $\Delta\alpha(t) = \alpha_{pc}(t) - \alpha_{ps}(t)$ as a function of Markov cycle. The excess grows from zero in the early cycles and saturates at a negative value ($|\alpha_{pc}| > |\alpha_{ps}|$) by mid-circuit, persisting to the final cycle. This rules out an early-time initialization artifact.

d. Test 4 (averaging order). Figure 10 compares two procedures: fitting each sample then averaging the slopes (“slope average”) vs. averaging the correlator curves then fitting (“average then fit”). The two give consistent values, ruling out a Jensen-inequality bias as the dominant source of the excess.

e. Test 5 (neutralized ensemble). A subset of samples with near-zero centered total charge ($|Q_{pct}| < \epsilon$) is selected, and the correlator slope is refit. The slope excess persists in this neutralized ensemble, indicating that charge-sector contamination from off-half-filling trajectories is not the primary driver.

C. Interpretation

The diagnostic battery collectively suggests that the slope excess is a genuine property of the perfect-correction protocol ensemble rather than a numerical or sampling artifact. A plausible physical mechanism is the following: under perfect correction, the feedback unitary applied at each cycle introduces a coherent phase shift that is *correlated* across the sample batch—the correction is derived from the same measurement outcomes that determined the post-measurement state. This correlation enriches the off-diagonal structure of the ensemble-averaged occupation matrix relative to the uncorrelated post-selected trajectory, producing a faster algebraic decay of $\tilde{C}^2(r_y)$.

An alternative explanation involves the boundary conditions on the charge sector: perfect correction enforces approximate $\hat{Q} \approx N_x N_y$ at the ensemble level, which may impose additional off-diagonal coherences across unit cells that the post-selected trajectory (which selects one charge sector deterministically) does not share. Distinguishing these mechanisms conclusively would require access to the full sample covariance matrix $\langle G_{\alpha\beta} G_{\gamma\delta} \rangle_S$, which is not currently saved.

VIII. PHYSICAL IMPLICATIONS

a. Rapid purification and topological ordering. The most striking observation from Campaign I is the speed of purification. Within $O(N_y)$ adaptive Markov cycles the system transitions from a maximally disordered maximix state to a near-pure Chern-insulator state with $|C_{RS}| \approx 1$ and charge variance $\ll N_x N_y / 2$. This is consistent with the theoretical expectation that adaptive circuits in the “area-law” phase of the MIPT can purify exponentially fast [8]. The convergence of the real-space Chern number confirms that the steady state selected by the adaptive feedback is topologically nontrivial—the measurement back-action drives the system toward a specific topological phase rather than a featureless Fock state.

b. Charge conservation and measurement-induced order. The $U(1)$ charge symmetry of the Class A circuit has a direct effect on the charge-variance dynamics. The rapid suppression of $\text{Var}(\hat{Q})$ from the maximix baseline is not merely purification (which would decrease all entropy sources equally) but reflects a specific *charge crystallization*: individual measurement trajectories localize in a narrow charge sector much more rapidly than a symmetry-breaking circuit would. The persistence of small residual charge variance ($\sim 10^{-2}$) at late times under perfect correction, compared to the even smaller residual under post-selection, quantifies how strongly the ensemble spread in charge deviates from the post-selected pure-state value.

c. Critical correlations in the domain-wall geometry. Campaign II reveals that in the presence of a topological domain wall, the circuit generates long-range algebraic correlations in the off-diagonal covariance $\tilde{C}^2(r_y) \propto r_y^\alpha$. The fact that the entropy grows logarithmically with chord length (consistent with Eq. (9) with finite c) and the correlator decays as a power law are hallmarks of a critical or weakly volume-law entangled state. Whether this rep-

resents true criticality (i.e., the circuit sits at or near a measurement-induced critical point) or a crossover between a bulk topological phase and edge-localized modes is not yet determined; a systematic N_y scaling analysis of c and α is needed to distinguish these scenarios.

d. Protocol dependence and its diagnostic implications. The systematic difference in correlator scaling between perfect correction and post-selection (Sec. VII) has implications for the interpretation of circuit simulations. If the slope excess is physical rather than artifactual, it implies that the measurement-averaged ensemble (perfect correction) and the individual trajectory (post-selection) belong to different universality classes of the MIPT, in agreement with the general principle that the MIPT is a property of the distribution of quantum states rather than any single trajectory. Verifying this interpretation requires larger system sizes and, ideally, access to the replica structure [5, 9] of the measurement-averaged state.

e. Future directions. Several extensions are immediately motivated: (i) completing the $N_y = 60$ streaming covariance run to extend the finite-size scaling of c and α ; (ii) computing the two-copy (second-Rényi) entropy to access the entanglement negativity spectrum and probe the MIPT universality class more directly; (iii) analyzing the spatial charge density profile near the domain wall to extract the localization length of the edge modes as a function of cycle; (iv) performing a finite-size scaling collapse of $\tilde{C}^2(r_y)$ to extract the critical exponent ν of the MIPT.

ACKNOWLEDGMENTS

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- [1] A. Nahum, J. Ruhman, S. Vijay, and J. Haah, *Quantum Entanglement Growth Under Random Unitary Dynamics*, Phys. Rev. X **7**, 031016 (2017).
 - [2] Y. Li, X. Chen, and M. P. A. Fisher, *Quantum Zeno effect and the many-body entanglement transition*, Phys. Rev. B **98**, 205136 (2018).
 - [3] B. Skinner, J. Ruhman, and A. Nahum, *Measurement-Induced Phase Transitions in the Dynamics of Entanglement*, Phys. Rev. X **9**, 031009 (2019).
 - [4] A. Chan, R. M. Nandkishore, M. Pretko, and G. Smith, *Unitary-projective entanglement dynamics*, Phys. Rev. B **99**, 224307 (2019).
 - [5] Y. Bao, S. Choi, and E. Altman, *Theory of the phase transition in random unitary circuits with measurements*, Phys. Rev. B **101**, 104301 (2020).
 - [6] A. Kitaev, *Anyons in an exactly solved model and beyond*, Ann. Phys. **321**, 2 (2006).
 - [7] P. Calabrese and J. Cardy, *Entanglement entropy and quantum field theory*, J. Stat. Mech. **2004**, P06002 (2004).
 - [8] M. J. Gullans and D. A. Huse, *Dynamical Purification Phase Transition Induced by Quantum Measurements*, Phys. Rev. X **10**, 041020 (2020).
 - [9] C.-M. Jian, Y.-Z. You, R. Vasseur, and A. W. W. Ludwig, *Measurement-induced criticality in random quantum circuits*, Phys. Rev. B **101**, 104302 (2020).